

An Inert Museum-Case Monitor

Compose hazards without pretending to control hardware

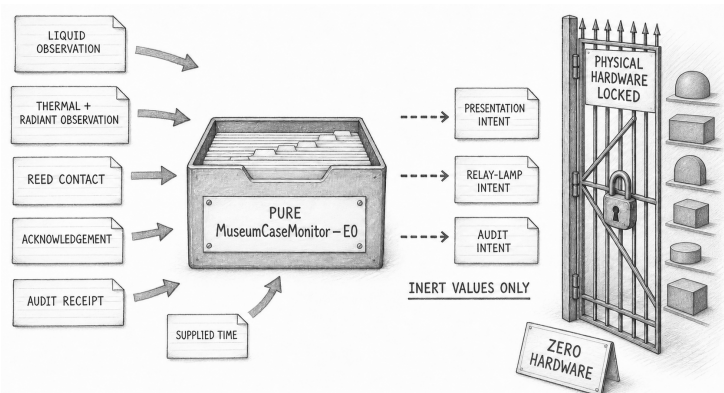
E0 SYNTHETIC REPLAY — NO SENSOR, RELAY, OR STORAGE HARDWARE

Lesson	063 — integrating project	Time	180–240 minutes
Prerequisites	Lessons 061–062 and magnetic observations	Board	compile-only Mega replay
Input	one complete copied hazard frame	Observation	named intent and audit cells

Responsibility. `MuseumCaseMonitor` composes copied liquid, thermal, radiant, reed-contact, acknowledgement, and recorder evidence. It owns a deterministic alarm/cooldown state machine, inert presentation and relay-lamp intent, and one bounded audit transaction. It does not sample a sensor, switch a relay, write storage, read an RTC, control access, or certify preservation or safety.

1 Draw the boundary around policy

The monitor begins with already-qualified copied observations. Lesson 061 provides liquid quality; Lesson 062 provides independent thermal and radiant quality; a copied magnetic observation preserves the complete reed-contact record. The operator acknowledgement and recorder receipt are copied evidence too. Supplied time is the only clock.



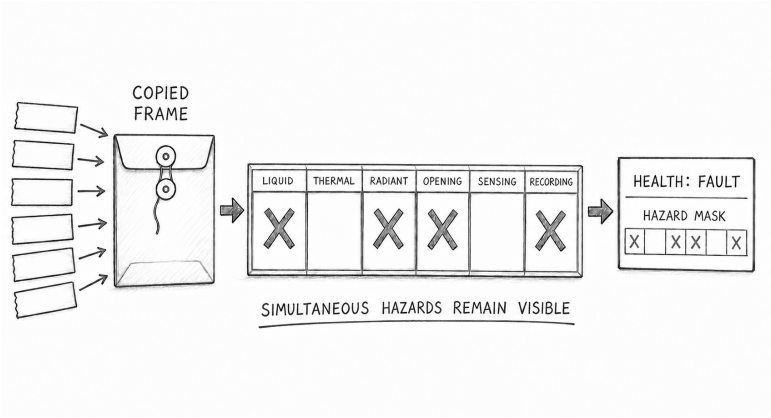
Alternative text: a pencil drawing feeds six copied evidence cards into an inert museum-case policy; a locked physical gate separates the policy from unidentified sensors, display, sounder, relay lamp, RTC, and storage, and no wire crosses the gate.

E0 proves	E0 does not prove
hazard precedence and cooldown from copied evidence	simultaneous physical sampling
inert RGB, LCD, sound, and relay-lamp intent	any energized output or safe electrical state
one bounded recorder intent/receipt transaction	durable storage or wall-clock accuracy
deterministic replay and attribution	fire detection, security, or preservation fitness

There is no formal electrical schematic at E0. An authoritative schematic requires every exact specimen, independent electrical qualification, resource and safe-state evidence, and measured combined acceptance. A generic relay or sensor drawing would conceal the open work rather than teach it.

2 Keep the complete hazard frame attributable

Every frame names the exact expected source and revisions fixed by `MuseumCaseConfig`. The liquid and environment values preserve their complete Lesson 061 and 062 observations. The reed wrapper preserves the complete qualified magnetic observation, not merely a Boolean. A qualified active contact means case closed; qualified inactive means reed-open and case-open, producing the **Opening** hazard.



Alternative text: a hand-drawn additive hazard ledger retains liquid, radiant, opening, and recording bits together; recording makes the reported health fault without erasing the other hazard evidence.

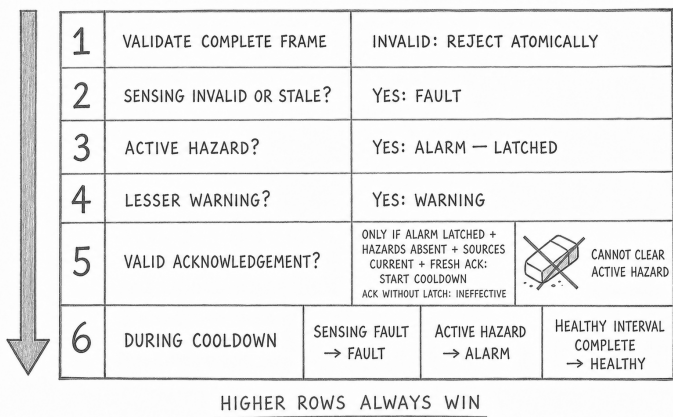
Structural validation precedes interpretation. Zero or duplicate role IDs, wrong revisions, unknown enum/status encodings, future or modularly ambiguous times, ordering violations, and mismatched receipt fields reject atomically. A convincingly alarming value cannot rescue an invalid frame. By contrast, a structurally valid but too-old required observation is admitted as evidence of a sensing problem: it commits **Fault** with the **Sensing** bit. Ordinary unsigned rollover with a wrap-valid age follows that same semantic age rule rather than being rejected merely because the counter wrapped.

Evidence	Why the monitor retains it
six producer sequences	audit witness attribution and replay
independent observation times and ages	delayed sources are not called simultaneous
reed polarity, raw level, quality, and edges	case-open meaning is not reduced to a bare bit
operator source, sequence, time, and status	acknowledgement remains attributable intent

Predict the result when the reed Boolean appears closed but its copied magnetic quality is faulted: health _____; sensing hazard present? _____.

3 Apply fault and hazard precedence

For a valid frame, lifecycle state first prevents active output intent. A sensing fault or stale required source produces **Fault**, never **Healthy**. Wet liquid, thermal alarm, radiant hazard, or case-open evidence latches **Alarm**. Lesser damp, uncertainty, or disagreement conditions produce **Warning**.



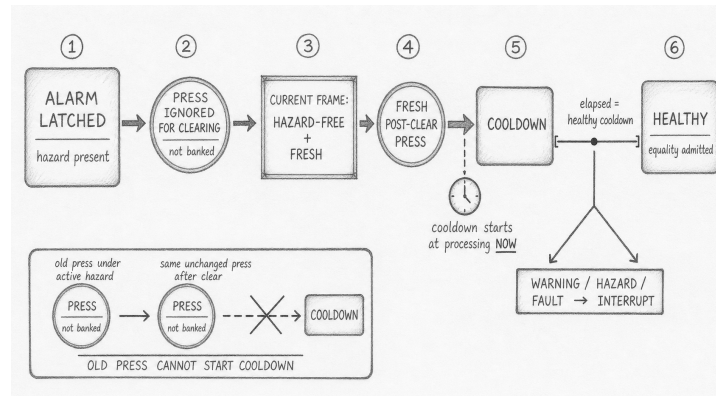
Alternative text: a pencil precedence ladder checks frame validation, sensing faults, active hazards, lesser warnings, the latched-alarm and fresh-acknowledgement condition, and cooldown interruption or completion.

Condition	Health	Hazard evidence retained
stale or invalid required sensing	Fault	Sensing plus other supported bits
wet, thermal alarm, radiant hazard, or open case	Alarm	every simultaneous hazard bit
damp, thermal warning, or disagreement	Warning	lesser condition remains attributable
first complete clean frame, with no prior alarm latch	Healthy	none
clean recovery from a latched alarm after acknowledgement	Cooldown	none
clean latched-alarm recovery after cooldown	Healthy	none

A single highest-alarm enum would lose collisions. The hazard mask therefore retains **Liquid**, **Thermal**, **Radiant**, **Opening**, **Sensing**, and **Recording** independently. If liquid and opening alarms arrive together, record the expected health _____ and hazard combination _____.

4 Acknowledge only after hazards clear

Acknowledgement records operator intent; it cannot clear an active hazard or sensing fault. Cooldown exists only to recover a previously latched alarm. After that alarm's hazards are absent and every required source is current, a fresh acknowledgement starts the configured cooldown. An acknowledgement when no alarm is latched is ineffective and cannot move a first clean frame away from **Healthy**. Any hazard or sensing fault during cooldown relatches alarm or fault. A lesser warning interrupts cooldown and reports **Warning**; a later clean frame still needs a fresh acknowledgement before recovery can restart. Only a complete clean cooldown returns the latched monitor to **Healthy**.



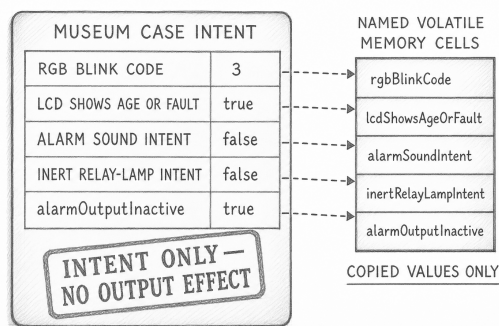
Alternative text: a pencil state trace moves from qualifying to alarm, ignores acknowledgement while a hazard remains, enters cooldown only after clean evidence and acknowledgement, interrupts on warning, relatches on a new hazard, and reaches healthy only after a restarted complete clean cooldown.

1. First complete clean frame with a pressed acknowledgement: _____.
2. Alarm plus pressed acknowledgement: health remains _____.
3. Clean current frame after that alarm plus a fresh press: next state _____.
4. Warning one tick before cooldown completion: next state _____.
5. Clean evidence exactly at the restarted cooldown boundary: admitted or delayed? _____.
6. Repeated ineffective acknowledgement with unchanged semantics: creates a new audit decision? _____.

initialize(now) advances the nonzero lifecycle generation and publishes qualifying, inactive outputs. **reset(now)** advances it again, invalidates old receipts, clears volatile retry/cooldown history, and returns to qualifying without claiming present hazards were cleared. **shutdown()** invalidates audit state and publishes canonical inactive outputs; repeated shutdown is idempotent.

5 Publish intent, not hardware effects

The output is a **MuseumCaseIntent**: health, complete hazard mask, grayscale-safe RGB blink code, LCD age-or-fault visibility, sound intent, inert relay-lamp intent, and an explicit inactive-alarm-output flag.

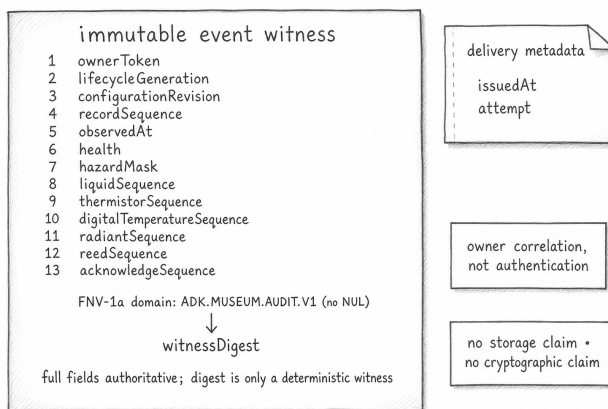


Alternative text: a pencil intent card copies RGB blink code, LCD age-or-fault visibility, alarm-sound intent, inert relay-lamp intent, and alarm-output-inactive into matching named volatile memory cells under an explicit no-output-effect stamp.

Shutdown and every not-initialized result require sound and relay-lamp intent false and **alarmOutputInactive** true. These values do not prove that a physical output became inactive. Future E1c/E1d work must prove the presentation endpoints and combined passive fixture; E2 may qualify only an extra-low-voltage, current-limited inert lamp behind an independently removable supply. The relay may never control mains, a lock, access, confinement, egress, or a safety load. RGB, LCD, and sound intent are diagnostic presentation, not security or life-safety signaling.

6 Freeze one audit witness at a time

Every record-creating semantic decision publishes at most one outstanding **MuseumAuditIntent**. Its immutable witness includes owner, lifecycle and configuration, record sequence, observation time, health and hazard mask, all six producer sequences, and a 32-bit FNV-1a digest. Delivery **issuedAt** and attempt are not part of the event witness.



Alternative text: a pencil audit card seals owner, lifecycle, configuration, record, observation time, health, hazard mask, and six source sequences beneath one witness digest, while issued-at and attempt sit on a replaceable delivery tab outside the seal.

The digest runs FNV-1a over ASCII **ADK.MUSEUM.AUDIT.V1** without its terminating NUL, followed by the canonical fields in fixed order. Integers are little-endian; enum and mask values use one byte. Full witness fields are always compared, so the digest never substitutes for the record.

The frozen test vector uses owner **0x01020304**, lifecycle **0x11223344**, configuration **0x5566**, record **0x778899aa**, observed milliseconds **0x0a0b0c0d**, health **Alarm**, hazard mask **0x19**, and producer sequences {1, 2, 3, 4, 5, 6}. Predict the frozen digest before checking the test: **0x_____**. The canonical answer is **4086e509**.

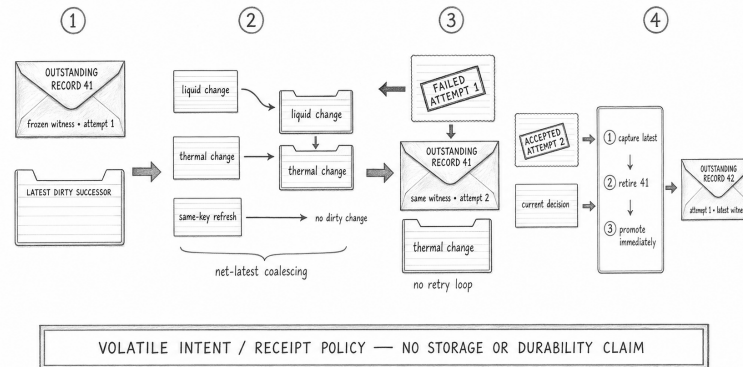
The first complete sensed decision after initialize or reset creates a record. A clean first frame is **Healthy**; depending on its evidence, the first decision may instead be **Warning**, **Alarm**, **Fault**, or evidence-driven **Qualifying**. Initial **Cooldown** is unreachable because no alarm has yet been latched. Later record creation uses the exact semantic key {health, hazard mask}. A fresh same-key sample updates live provenance but does not manufacture a record. Every health or hazard-mask change does.

7 Retain the latest successor during delivery

The monitor has one outstanding immutable record and one **dirtySuccessor** slot. If decisions change while delivery is pending, the outstanding witness remains frozen and the slot retains the latest net decision. Several changes may coalesce only to that latest bounded successor. The slot obeys three complete rules on every valid frame:

1. if the current semantic key returns to the outstanding record's key, clear the dirty slot because there is no net successor;
2. if the key still matches the dirty slot, refresh the slot's complete current witness fields, including the latest producer sequences and observation time;
3. otherwise replace the slot with the complete current decision.

The slot has no record sequence, digest, attempt, or deadline until promotion.



Alternative text: a pencil sequence keeps audit record 41 sealed while liquid then thermal changes replace one latest dirty successor; a failed first attempt retains both, and an accepted attempt-two receipt retires 41 and immediately promotes the latest card as record 42.

A receipt must match the owner, lifecycle, configuration, record sequence, every witness field, digest, and current attempt. Acceptance retires the record. Failure or deadline loss adds **Recording**, increments the bounded attempt, and later reissues the same witness. There is no retry loop. The inclusive deadline remains pending; the next valid tick records loss.

When an accepted receipt and a changed decision arrive together, validation and latest-successor capture occur first. Then the receipt retires the old record and the successor is immediately assigned the next sequence, digest, deadline, and attempt one. This order prevents the newest decision from being silently treated as covered by an older receipt.

Collision	Required result
current key equals outstanding key	clear any dirty successor
current key equals dirty key	refresh its complete latest witness
current key differs from both	replace with complete current decision
several key changes while pending	one net-latest complete successor retained
failed receipt with successor present	immutable record and successor both retained
accepted receipt plus current change	retire old; immediately publish latest successor
final attempt fails	freeze both, assert recording fault, emit no retry

Only explicit reset or shutdown may discard terminal volatile audit state; that discard is never reported as storage success. RTC accuracy, media durability, atomic file commit, wear, and recovery after power loss remain deferred physical work.

8 Replay the complete collision trace

The Mega sketch feeds copied result cells into one monitor. The caller owns the result buffer passed to **update**; the monitor neither allocates nor returns a hidden aggregate. The sketch does not call storage or acquire hardware.

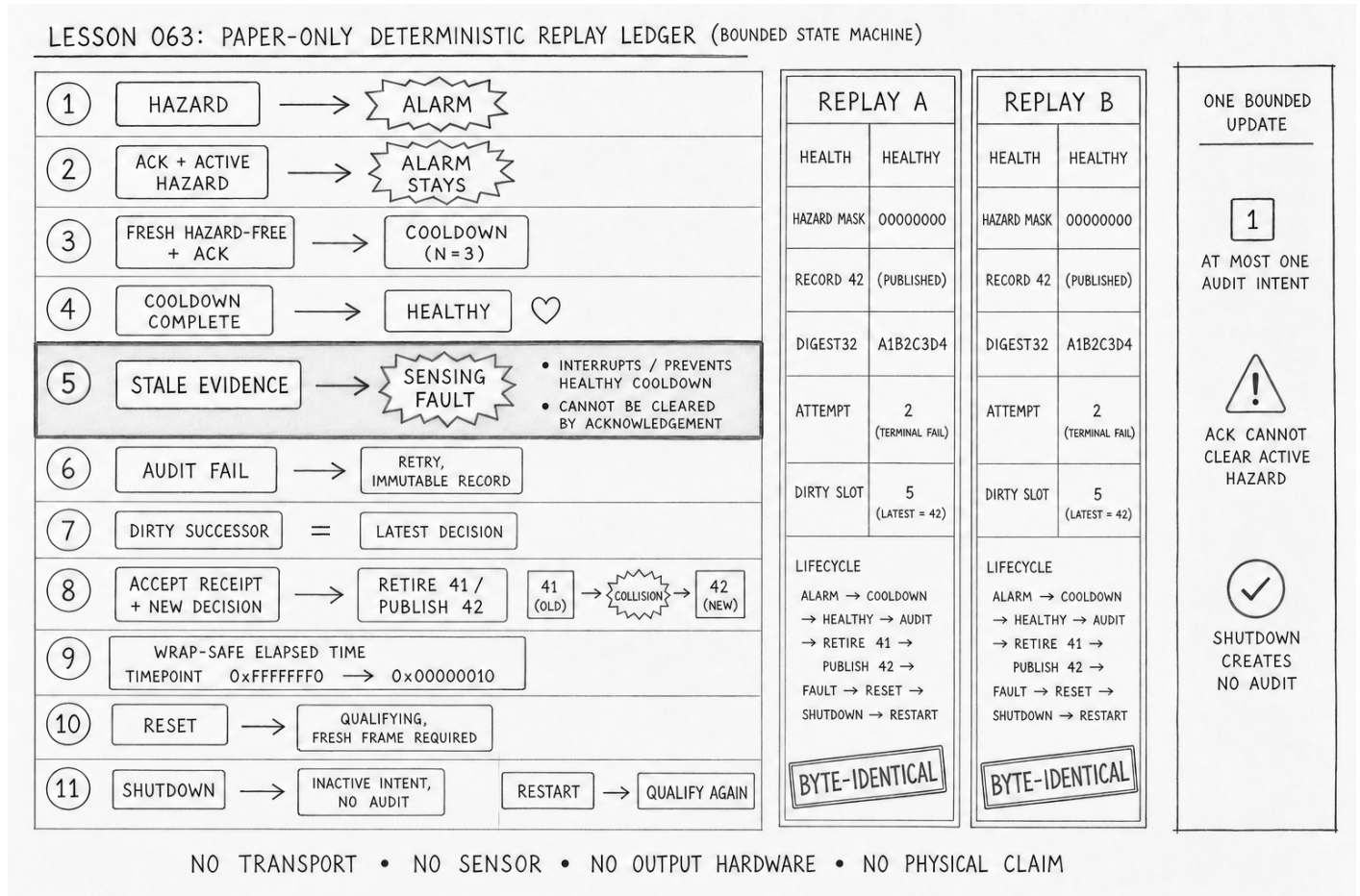
```
MuseumCaseMonitor monitor (config);
monitor.initialize (TimePoint (0));

MuseumCaseResult result = {};
result.intent = monitor.snapshot ();
const Status updateStatus = monitor.update (frame, result);
if (updateStatus.ok () && result.hasAuditIntent) {
    auditIntentResult = result.auditIntent;
}
```

```

if (updateStatus.ok ()) {
    intentResult = result.intent;
}

```



Alternative text: a pencil replay ledger covers first complete decisions, simultaneous hazards, stale sensing, ineffective acknowledgement, cooldown, record 41, failed delivery, changing successor, accepted receipt, record 42, terminal retry exhaustion, reset, rollover, and shutdown in named result cells.

The canonical collision starts with alarm record 41, changes liquid and thermal evidence during attempt one, fails that attempt, changes radiant and case-opening evidence during attempt two, then accepts attempt two in the same update as acknowledgement changes. Record 42 must contain the latest six producer sequences and hazard state, not either intermediate witness.

Complete these diagnoses:

1. Wrong digest but otherwise matching receipt: _____.
2. Matching receipt one tick after inclusive deadline: _____.
3. Accepted retired receipt repeated byte-for-byte: _____.
4. Record sequence would wrap to zero: _____.
5. Shutdown with an outstanding record: output becomes _____; storage success claimed? _____.

9 Inspect gates and record evidence

Gate	E0 evidence	Still open
composition	exhaustive hazard and cooldown replay	simultaneous powered sensing
outputs	inert intent and inactive lifecycle results	endpoint acquisition and safe electrical state
audit	witness, collision, receipt, retry, and reset traces	durable media and wall-clock behavior
resources	exact compiler and object/stack gate	physical current and combined rail budget
observation	named volatile result cells	RGB/LCD/sound/relay physical evidence

The ordinary Mega replay measures 21,348 bytes of flash and 1,405 bytes of static SRAM. The isolated no-LTO resource gate measures 22,980 bytes of flash, 1,405 bytes of static SRAM, and 819 bytes of conservative synchronous stack. Aggregate policy

objects occupy 709 bytes: 528 for the monitor, 69 for the Lesson 061 policy, and 112 for the Lesson 062 policy. The caller-owned envelope and result buffers occupy 207 and 72 bytes respectively; each passes the 256-byte per-buffer limit, while their combined footprint is 279 bytes. After the specified reserve and stack accounting, 5,840 bytes of Mega SRAM remain.

The scoped evidence fingerprint is `d376c43b63f42f2ed48c768d9afcaa2583d5a2b4e194a45f9a3f125694e41543`. Every flash, SRAM, stack, object, caller-buffer, residual, ownership, and hidden-return hard gate passes; the 819-byte stack result is an independently reviewed target miss below the 896-byte hard limit. The ordinary sketch and isolated no-LTO probe are separate measurement boundaries; their flash and static-SRAM numbers must not be added.

Physical gates remain open. E1c qualifies the exact reed and harmless presentation endpoints. E1d proves the combined passive monitor, including current, rail, source age, disagreement, shutdown, and non-Serial observation. E2 may qualify only an extra-low-voltage, current-limited inert relay/lamp fixture with independent power removal. No mains, flame, heater, ignition, laser, lock, access, confinement, egress, preservation, or life-safety function is authorized.

Completion check	Result or evidence
Exact identities, nested provenance, and atomic frame validation	
Every health transition and hazard-mask add/remove combination	
Acknowledgement, cooldown boundaries, relatch, reset, and shutdown	
Frozen witness vector, digest, receipt matching, and deadlines	
Dirty successor, collision order, retry exhaustion, and rollover	
Canonical resource gate and reviewed target dispositions	
Zero-hardware E0 boundary and all E1c/E1d/E2 gates	

Completion statement: I can preserve simultaneous and attributable hazards, explain why acknowledgement cannot clear active evidence, replay the bounded audit transaction and collision order, and identify every physical claim that remains open.

Learner: _____ Reviewer: _____ Date: _____

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